

Aarhus University

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Parallax - Requirement Specification

Students:

Thea Teresa Lindgaard (202204350)
Niels Jakob Østerberg (202105879)
Kasper Stecher (202100451)
Kasper Bentsen (202103249)

Counsellor:

Lukas Esterle

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1. Actors

For this project, some actors have been identified. A short description of each actor can be seen in the following sections. The actors have been separated into categories.

1.1. Game Actors

In the following sections, “player” will be used to reference both player types, and a specifier “Human” or “Cat” will be added when something applies to only a specific player type.

Name	Player (Human)
Type	Primary
Description	A person who wants to play a game with a friend, as the human character

Name	Player (Cat)
Type	Primary
Description	A person who wants to play a game with a friend, as the cat character

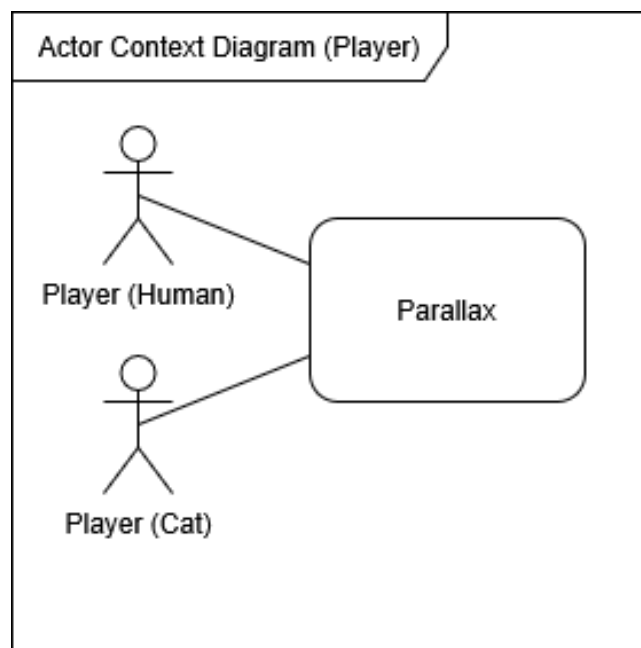


Figure 1: Actor Context Diagram.

1.2. Network Actors

In the following sections, “Host” will be used to reference the player hosting the game lobby, and “Client” will be used to reference the player joining the hosted game.

Name	Network Host
Type	Primary
Description	Person hosting the game session

Name	Network Client
Type	Primary
Description	Person joining the game session

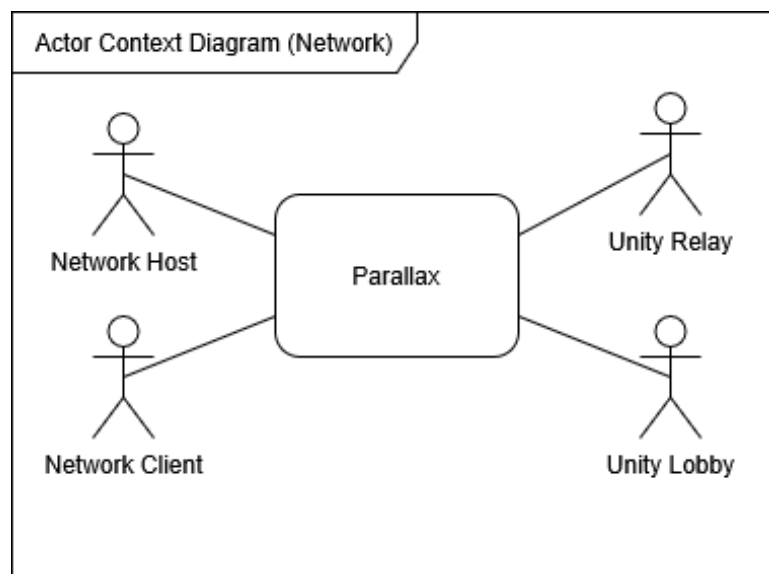


Figure 2: Network Actor Context Diagram.

2. User Stories

In the following sections you can find the user stories belonging to each epic. The user stories will serve as the functional requirements for this project.

User Story List

1	E1-US1: Player Movement	4
2	E1-US2: Player (Cat) Jumping	4
3	E1-US3: Player Collision	4
4	E1-US4: Player Camera	4
5	E2-US1: Realtime Visual Feedback	4
6	E2-US2: Pick Up Items	4
7	E2-US3: Push/Pull Obstacles	5
8	E2-US4: Asymmetric Perspective	5
9	E2-US5: In-Game Communication	5
10	E2-US6: In-Game Voice Communication	5
11	E3-US1: Level Timer	5
12	E3-US2: Time Trial	5
13	E3-US3: Level Failure	6
14	E3-US4: Level Failure Restart	6
15	E3-US5: Player Life System	6
16	E3-US6: Damage from Hazards	6
17	E4-US1: Join Lobby	6
18	E4-US2: Create Lobby	6
19	E4-US3: Role selection	7
20	E4-US4: Reconnect	7
21	E4-US5: Host Leave Level	7
22	E4-US6: Client Leave Level	7
23	E5-US1: Main Menu Navigation	7
24	E5-US2: In-game Menu	7
25	E5-US3: Settings Menu	8
26	E5-US4: Guide Menu	8
27	E6-US1: Level Progression	8
28	E6-US2: Auto Save	8
29	E6-US3: Load Game Progress	8
30	E6-US4: Level Restart	8

2.1. Epic 1: Core Controls

E1-US1: Player Movement
As a player I want directional movement So I can navigate the game environment horizontally

E1-US1: Player Movement

E1-US2: Player (Cat) Jumping
As player (cat) I want to jump So I can navigate the game environment vertically

E1-US2: Player (Cat) Jumping

E1-US3: Player Collision
As a player I want collision detection So I cannot walk through obstacles

E1-US3: Player Collision

E1-US4: Player Camera
As a player I want a camera that follows my character So I see the game world

E1-US4: Player Camera

2.2. Epic 2: Cooperative Game Mechanics

E2-US1: Realtime Visual Feedback
As a player I want to be able to see the other player in real-time So we can cooperatively solve the game challenges

E2-US1: Realtime Visual Feedback

E2-US2: Pick Up Items
As a player (human) I want to pick light-weight objects up So i can solve puzzles

E2-US2: Pick Up Items

E2-US3: Push/Pull Obstacles
As a player (human) I want to push or pull heavy obstacles around So i can solve puzzles

E2-US3: Push/Pull Obstacles

E2-US4: Asymmetric Perspective
As a player I want to observe the game environment in a way exclusive to my role So that each player has a separate set of objectives

E2-US4: Asymmetric Perspective

E2-US5: In-Game Communication
As a player I want to be able to give the other player a visual and audible clue So I can communicate my intentions

E2-US5: In-Game Communication

E2-US6: In-Game Voice Communication
As a player I want to be able to use in-game voice communication to interact with the other player So we can coordinate our actions

E2-US6: In-Game Voice Communication

2.3. Epic 3: Core Game Mechanics

E3-US1: Level Timer
as a player I want a timer to track my time used on a level So that I can improve my best time

E3-US1: Level Timer

E3-US2: Time Trial
As a player I want levels to have a time goal So the game provides a challenge

E3-US2: Time Trial

E3-US3: Level Failure
As a player I want to fail the level when I make critical mistakes So the game provides challenge and consequence

E3-US3: Level Failure

E3-US4: Level Failure Restart
As a player I want the level to restart when I fail So I can try again

E3-US4: Level Failure Restart

E3-US5: Player Life System
As a player I want my character to have health So I can be challenged and avoid taking damage

E3-US5: Player Life System

E3-US6: Damage from Hazards
As a player I want to take damage when I interact with dangerous objects So I am challenged to avoid hazards

E3-US6: Damage from Hazards

2.4. Epic 4: Networking and Multiplayer

E4-US1: Join Lobby
As a client I want to be able to join a lobby So I can play with a host

E4-US1: Join Lobby

E4-US2: Create Lobby
As a host I want to be able to create an online lobby So I can have a client join

E4-US2: Create Lobby

E4-US3: Role selection
As a player I want to choose my role So I can play the chosen role

E4-US3: Role selection

E4-US4: Reconnect
As a host or client I want to reconnect to a session after a disconnect So I do not lose progress and won't have to start over

E4-US4: Reconnect

E4-US5: Host Leave Level
As a host I want to be returned to the lobby when I leave a level So I can continue playing with the same client

E4-US5: Host Leave Level

E4-US6: Client Leave Level
As a client As a client, I want to return to the main menu when I leave a level So I can start a new game

E4-US6: Client Leave Level

2.5. Epic 5: Application Navigation and Settings

E5-US1: Main Menu Navigation
As a player I want a main menu with navigation options So I can access game, settings and exit

E5-US1: Main Menu Navigation

E5-US2: In-game Menu
As a player I want an in-game menu So I can return to the main menu

E5-US2: In-game Menu

E5-US3: Settings Menu
As a player I want a settings menu So I can adjust the game settings

E5-US3: Settings Menu

E5-US4: Guide Menu
As a player I want a guide menu So I can lookup the game interactions

E5-US4: Guide Menu

2.6. Epic 6: Game Progression

E6-US1: Level Progression
As a player I want to complete levels with the help of another player So I can progress in the game

E6-US1: Level Progression

E6-US2: Auto Save
As a player I want the game to automatically save my progression after a completed level So I can continue my progression later

E6-US2: Auto Save

E6-US3: Load Game Progress
As a player I want to be able to load my saved game So I can continue my progression

E6-US3: Load Game Progress

E6-US4: Level Restart
As a player I want to be able to reload the ongoing level So I can try the same level if I get stuck.

E6-US4: Level Restart

3. MoSCoW

The following table describes the **MoSCoW** classification for each user story.

Functional Requirements	MoSCoW
E1-US1: Player Movement	Must
E1-US2: Player (Cat) Jumping	Must
E1-US3: Player Collision	Must
E1-US4: Player Camera	Must
E2-US1: Realtime Visual Feedback	Must
E2-US2: Pick Up Items	Must
E2-US3: Push/Pull Obstacles	Must
E2-US4: Asymmetric Perspective	Must
E2-US5: In-Game Communication	Could
E2-US6: In-Game Voice Communication	Wont
E3-US1: Level Timer	Could
E3-US2: Time Trial	Could
E3-US3: Level Failure	Should
E3-US4: Level Failure Restart	Must
E3-US5: Player Life System	Should
E3-US6: Damage from Hazards	Could
E4-US1: Join Lobby	Must
E4-US2: Create Lobby	Must
E4-US3: Role selection	Could
E4-US4: Reconnect	Should
E4-US5: Host Leave Level	Should
E4-US6: Client Leave Level	Should
E5-US1: Main Menu Navigation	Must
E5-US2: In-game Menu	Must
E5-US3: Settings Menu	Could
E5-US4: Guide Menu	Could
E6-US1: Level Progression	Should
E6-US2: Auto Save	Could
E6-US3: Load Game Progress	Could
E6-US4: Level Restart	Should

3.1. Non-Functional Requirements

The following section describes the non-functional requirements and their **MoSCoW** classifications.

Usability	MoSCoW
A player is able to start a lobby in less than 3 clicks	Should
The game is fully navigable using a keyboard and mouse	Must
The game is fully navigable using a controller	Could
The game provides a color-blind accessibility mode	Wont
Game mechanics are introduced gradually	Should

Reliability	MoSCoW
Player positions are synchronized with no desync longer than 500 ms	Should
The game automatically tries to reconnect players for 30 seconds, after a connection loss	Could
The game will direct a player to the main menu, if a connection loss lasts more than 30 seconds	Should

Performance	MoSCoW
The game maintains at least 30 FPS on the target platform	Should
Initial game load time is under 30 seconds	Could
Network latency does not introduce more than 150 ms input delay	Should
CPU usage should not exceed 80% on the target platform during peak gameplay	Could
Scene transitions must complete within 5 seconds	Should

Supportability	MoSCoW
The game can receive updates post-release	Wont
Players are able to report bugs	Wont
The game monitors diagnostics during gameplay	Wont

Security	MoSCoW
The server validates all client inputs	Could
User data complies with GDPR regulations	Wont
The system must prevent basic client-side cheating by validating all gameplay-critical actions server-side	Could

3.2. System Specifications

The following section outlines the minimal hardware and networking available to the group for testing purposes. This is the baseline for which each of the non-functional requirements are measured.

3.2.1. Platform

To ensure a consistent reference for measuring frame-rate and performance, the target platform was chosen as the team member's PC with the following specs:

RAM	32GB
RAM frequency	5600 MHz
CPU	Intel(R) Core(TM) Ultra 7 155U
GPU	Intel(R) Graphics (integrated)
Wi-Fi Card	Intel(R) Wi-Fi 6E AX211 160MHz
Screen Resolution	1920 x 1200 px

3.2.2. Network

To ensure a consistent reference for multiplayer testing and synchronization, the following network conditions were used during development and testing:

Connection Type	Home broadband connection (Wi-Fi/Ethernet)
Target Bandwidth	Minimum 25 Mbps download / 5 Mbps upload
Target Latency	Below 100 ms average ping between players
Maximum Accepted Input Delay	150 ms
Packet Loss Tolerance	Up to 2% packet loss without disconnect
Target Player Count	2 concurrent players
Network Architecture	Host-authoritative client-host model using Unity Netcode for GameObjects